

Precision 5520 → GPD Pocket 3

What is on the daily driver, and what the stick puts on the Pocket 3

2026-09-02

Legend: **yes** = the setup script installs or restores it, unchanged. **no** = deliberately left off the Pocket 3. **by hand** = cannot be scripted; see the manual checklist.

Everything in the *Laptop* column was read from the live Precision 5520 on 2026-09-02. Everything in the *Pocket 3* column describes what `setup-pocket3.sh` on the Ventoy stick does.

0. Distro verdict for the GPD Pocket 3

Verdict: Ubuntu 24.04 LTS, with the `umpc-ubuntu` quirks applied `sway-native`. Not because Ubuntu has the best Pocket 3 support in the abstract — Arch arguably does — but because on this device the support is *the same five kernel/udev quirks on every distro*, and only one choice also makes the machine a copy of the daily driver.

Distro	Pocket 3 support	Drawbacks	Copy fit
Ubuntu 24.04 (any flavour)	wimpysworld/ <code>umpc-ubuntu</code> is <i>the</i> reference tweak set; Ubuntu MATE built its official Pocket 3 image from it. Kernel 6.8 covers both CPU variants.	No official Pocket 3 image since Ubuntu MATE 21.10 — you apply the script (or these five files) yourself. Script is Xorg-first; its README says Wayland “mileage may vary”. 21.10 is EOL since 2022. Kernel 5.13: audio broken on the N6000, needs the HDA override on the 1195G7.	yes identical packages, PPAs, scripts
Ubuntu MATE 21.10 image	Only true out-of-the-box image ever shipped.	21.10 is EOL since 2022. Kernel 5.13: audio broken on the N6000, needs the HDA override on the 1195G7.	no dead release
Arch	Best-written wiki page for this device; rolling kernel gets fixes first (N6000 audio landed in 6.1).	You maintain it. Not what the 5520 runs. ArchWiki was unreachable for this research (Anubis bot-wall) — its claims here are from search snippets, not read at source.	no package drift
Fedora	Tested alongside Ubuntu 21.10 by Liliputing (2022): wifi, touch, pen, rotation fine; audio needed the same driver-config workaround; webcam choppy.	Same quirks, different package names — nothing from manifests/ transfers.	no
NixOS	<code>nixos-hardware</code> has GPD Pocket profiles (Pocket 1/2; Pocket 3 via community).	Entire config model differs from the 5520.	no
Kali (defencore repo)	Working rotation/touch/TLP configs for 2021.4.	Its own ToDo lists sound, stylus, multi-touch and sleep as unsolved.	no

Why sway makes Ubuntu the easy case. The only place the distros genuinely differ is how much display-server glue they need. Under Xorg the portrait panel needs a monitor section, a touch transformation matrix, an accelerometer daemon and a display-scaler script. Under sway it is output DSI-1 transform 90 scale 1.5 plus map_to_output for touch — the Wayland caveat in umpc-ubuntu’s README is about their Xorg tooling, not about the hardware.

What no distro fixes (as of this research, 2026-09-02):

- **Audio depends on which Pocket 3 you have.** i7-1195G7: SOF has no topology, needs dsp_driver=1. N6000: SOF works from kernel 6.1 — with reports of popping and volume resetting on reboot.
- **Webcam** is reported choppy; video recording failed in the Liliputing test. Not re-verified since.
- **Fan** runs on BIOS curves; no userspace control exists.
- **Suspend** is s2idle only. Deep sleep is not available; expect drain while “asleep”.
- **The installer runs rotated 90°.** Every distro. It is only the live session.
- **Fingerprint reader:** no report of it working on Linux was found.

Evidence quality: web pages and upstream config files, read on 2026-09-02, except ArchWiki (blocked). The Liliputing test is from 2022 on 21.10/Fedora 35; nothing more recent with the same rigour turned up.

1. Hardware

	Precision 5520 (laptop)	GPD Pocket 3
CPU	Intel i5-8350U, 4c/8t	i7-1195G7 (Tiger Lake) or Pentium N6000 (Jasper Lake) — the script detects which
RAM	23 GiB	16 GB (soldered)
Storage	954 GB NVMe: 192 G / + 736 G /data	your unit’s M.2
Display	eDP-1, landscape	8” 1200×1920, physically portrait, mounted sideways
Effective desktop	native	1920×1200 rotated, scaled ×1.5 → 1280×800
Touch / stylus	none	Goodix GXTP7380 touch + pen — needs a rotation matrix
Suspend	works	s2idle only; deep sleep unavailable
Webcam	works	reported choppy on Linux; capture may fail
Fan control	BIOS	BIOS only; no userspace control

2. OS and desktop

	Laptop	Pocket 3	
Distribution	Ubuntu 24.04.4 LTS (noble)	Ubuntu 24.04.4 LTS — same point release, same ISO era	yes
Kernel	6.8.0-50-generic	6.8 generic (stock). Liquorix headers not carried over	no
Session	sway 1.9 on Wayland	sway 1.9 on Wayland	yes
Status bar	custom status.sh (volume / brightness / wifi / battery / disk / mem / cpu / clock)	identical file	yes
Theme	safety-yellow #FFE600 focus, #B8A500 inactive, #666600 unfocused	identical	yes

	Laptop	Pocket 3	
Title font	Ubuntu Bold 8 · bar: JetBrains Mono 8	identical (fonts in the apt list)	yes
Wallpaper	photo PXL_20220525_111527965 .MP.jpg (6.5 MB)	same image, path rewritten to ~/.local/share/backgrounds/wallpaper.jpg	yes
Terminal	foot — black bg, white fg, alpha 0.92	identical foot.ini	yes
Launchers	wofi (Mod+d), bemenu bdmenu_ada (Mod+Shift+s)	identical	yes
Notifications	mako (systemd user unit)	mako installed; unit enabled on first sway start	yes
Audio	PipeWire + wireplumber	PipeWire + wireplumber, plus the variant-specific HDA override	yes
Login shell	bash	bash	yes
Display config	none needed	output DSI-1 transform 90 scale 1.5 + touch/pen map_to_output	new

3. Software — the numbers

Layer	Laptop	Pocket 3	Left out
apt, manually installed	427	412	15 (listed in §4)
snap apps (bases excluded)	20	16	anbox, whatsapp-4linux, whatsie, prusa-slicer-snap
flatpak apps	4	4	—
pip user packages	34	32	2 build artefacts
third-party apt sources	17 files (10 live, 7 .distUp-grade leftovers)	the 10 live ones	jammy/focal PPAs are tried and dropped if they 404 on noble
signing keyrings	6	6	—

4. What is *not* going on the Pocket 3

apt (15): grub-pc grub-pc-bin os-prober memtest86+ — BIOS-boot leftovers the Pocket 3 (UEFI) has no use for · linux-headers-6.8.0-50-generic linux-headers-liquorix-amd64 — kernel-pinned · steam-launcher steam-libs-amd64 steam-libs-i386 · virtualbox-7.0 qemu-system-x86 qemu-utils virt-manager swtpm ovmf

snap (4): anbox · whatsapp-4linux · whatsie (ferdium stays, so you keep one WhatsApp) · prusa-slicer snap 2.7.4 (the flatpak 2.9.4 stays)

Everything else — FreeCAD, KiCad, PrusaSlicer, BambuStudio, Betaflight, INAV, ExpressLRS, Arduino, esptool, gpredict, gqrx, noaa-apt, LibreOffice, Thunderbird, Brave, Chromium, Firefox, Chrome, VS Code, Zotero, Spotify, Discord, Teams, Ferdium, Audacity, LMMS, VLC, Wireshark, nmap, TeamViewer, Rust-Desk, Remmina, Syncthing, Docker, wine-staging, Lutris, TeX Live, pandoc, Doom Emacs, neovim, micro, ranger, btop, tmux, fzf, ripgrep — goes.

5. Config and dotfiles — file by file

File	Laptop	Pocket 3	
~/.config/sway/config	301 lines	same + 20-line Pocket 3 block appended	yes
~/.config/sway/status.sh	63 lines	identical	yes
~/.config/foot/foot.ini	colours + alpha	identical	yes
~/.config/gtk-3.0/settings.ini	present	identical	yes
~/.config/doom/{config,init,packages} + cheatsheet	present	identical; doom sync rebuilds the 423 MB of packages	yes
~/.bashrc additions	PATH, EDITOR=emacs -nw, emacsclient alias, nvm hook, opencode	appended verbatim, /home/adam → \$HOME	yes
~/.gitconfig	PATH, go() asdfgh0318 / gh credential helper	identical	yes
~/.claude/settings.json + CLAUDE.md	present	identical	yes
~/.claude/.credentials.json	present	not copied — log in again	by hand
~/.ssh/id_ed25519	present	not copied — bring on encrypted media	by hand
~/.config/gh/hosts.yml	present	not copied — gh auth login	by hand
dconf (1292 keys)	present	not restored — GNOME-only, unused under sway	no
crontab (2 jobs)	adam_laptop auto-update, vibecoding-logger	not installed; paths in adam_laptop are now fixed so you can	no

6. Custom tools

Tool	Laptop	Pocket 3	
go / go-layout	~/.local/bin/go-layout → VIBECODING/go-layout	own repo bundled; install.sh links it and adds go()	yes
bdmenu_ada	~/bin symlink	bundled + linked	yes
settings, -hub, -dmenu, -wofi, printerctl	~/bin symlinks → settings-tools	bundled + linked	yes
archangel	~/bin symlink → ŻYCIE/archangel	bundled + linked	yes
fcmpc	→ VIBECODING/fcmpc (FreeCAD MCP)	inside ŻYCIE tracked content	yes
research_tree	→ PRACA/Shroud_Comparison/...	inside ŻYCIE tracked content, not linked	partial
litlib	real file in ~/bin	not bundled	no
~/.local/bin (claude, codex-acp, mmdc, yt-dlp, httpx...)	24 entries	reinstalled by their package managers where possible	partial

7. Toolchains

	Laptop	Pocket 3	
node	22.22.1 via nvm	nvm → node 22 LTS, default alias	yes
npm / pnpm	10.9.4 / 10.29.1	corepack pnpm@latest	yes
python	3.12.3 + 34 user pkgs	3.12 (noble) + pip list	yes
uv	0.11.17	astral installer, latest	yes
java	OpenJDK 21.0.12	from apt list	yes
docker	29.1.3 + compose v2	from apt list, user added to docker group	yes
emacs / doom	29.3	29.3 + doom clone + sync	yes
nvim	0.9.5	0.9.5	yes
rust / go / cargo	not installed	not installed	—

8. Data

	Laptop	On the stick	
ZYCIE, tracked in zycie.git	990 MB, 800 files	1.8 GB shallow clone (working tree + 1-deep history)	yes
ZYCIE .git full history	5.2 GB	not carried; git fetch --unshallow later if you want it	no
ZYCIE, untracked	~47 GB	not carried	by hand
└ Papers	1.3 GB	rsync / Syncthing	by hand
└ ardupilot	1.9 GB	re-clone upstream	by hand
└ Applications	1.1 GB	not needed	no
└ Pulpit	637 MB	rsync	by hand
└ Zotero	112 MB	Zotero sync	by hand
└ org-mode	109 MB	rsync	by hand
└ Sync / HUB	217 + 217 MB	Syncthing re-pair	by hand
~/snap	9.9 GB	regenerated by snaps	—

9. What the stick holds

```

/dev/sda 30.6 GB SanDisk Cruzer, Ventoy 1.1.17, GPT, Secure Boot capable
├─ ubuntu-24.04.4-desktop-amd64.iso ~6 GB sha256 3a4c9877...99d1e
├─ PAYLOAD/ 1.8 GB 1084 files
│   └─ setup-pocket3.sh 12 idempotent phases
│   └─ README.md · MANUAL-CHECKLIST.md · this document
│   └─ manifests/ apt 412 · snaps 16 · flatpaks 4 · pip 32 · 10 sources · 6 keys
│   └─ dotfiles/ sway · foot · gtk · doom · gitconfig · bashrc additions · claude
│   └─ pocket3/ grub · alsa · touch udev · accel hwdb · sway output block
│   └─ wallpaper/ the jpg
├─ zycie/ tracked ZYCIE + go-layout + bdmnu_ada + settings-tools + archangel
└─ adam_laptop/ existing repo, /home/adam-koszalka paths fixed

```

10. Only on the Pocket 3

These five files exist because of the hardware and have no counterpart on the laptop. Values are from wimpysworld/umpc-ubuntu, the reference Ubuntu MATE's official Pocket 3 image is built from.

File	Does	Note
/etc/default/grub.d/gpd-pocket3.cfg	fbcon=rotate:1, video=DSI-1:panel_orientation=right_side_up, mem_sleep_default=s2idle, GFXMODE=1200x1920x32	boot console, plymouth, suspend
~/.config/sway/config (appended)	output DSI-1 transform 90 scale 1.5 + touch/pen map_to_output DSI-1	if upside down: 90 → 270
/etc/udev/rules.d/99-gpd-pocket3-touch.rules	libinput matrix 0 1 0 -1 0 1 for touch + stylus	taps land where you touch
/etc/udev/hwdb.d/61-gpd-pocket3-sensor-local.hwdb	ACCEL_MOUNT_MATRIX=-1,0,0;0,1,0;1,0,0	auto-rotate
/etc/modprobe.d/alsa-gpd-pocket3.conf	snd-intel-dspcfg dsp_driver=1	i7-1195G7 only ; N6000 must not get it

The transform 90 value could not be tested without the device. It is the single most likely thing to need a one-character change.